b) Breather

The breather is a small vessel. It is connected between conservator and air outlet. It contains silica gel. It is a dehydrating agent. The moisture in the incoming air is removed. The colour of the silica-gel is **blue** and **pink,** when wet or damp.

c) Explosion Vent

In the event of an accidental internal short circuit in the transformer, an arc is formed between the turns of the winding. Heat is produced by the arc. Due to this, a large volume of gas is produced. Provision must be made for rapid release of gas. Otherwise high pressure will be built up inside leading to the lip of the tank blown off. For this reason an explosion vent is provided on the tap of the tank. Under normal conditions air is not allowed to come in contact with the oil. Under short circuit conditions, the diaphragon is ruptured due to high pressure. The gas is expelled to atmosphere. If the high pressure gas releasing a portion of the hot oil may get splashed and cause injury to the workers in the transformer yard. The explosion vent's mouth is covered by a glass or aluminium.

d) Buchholz Relay

This is a device which is attached to an oil immersed transformer. It is fitted in the pipe connecting the transformer tank with the conservator.

It consists of two floats as shown in Fig. 6.14.

Two pairs of electrical contacts are provided. These contacts may get short circuited under certain situations.

When an insulation breaks down in a transformer, gas is generated in the oil. Quick generation of this gas leads to a serious fault. The gas rushes through the pipe and pushes the lower float to the right. The two lower contacts bridge together and closes to trip the circuit of circuit breaker. Now the transformer is disconnected from the supply. If the fault develops slowly, gas will also generated slow. This may not be sufficient to move the lower float. This gas gets collected gradually in the top of the relay chamber. The oil level gets lowered. This causes the upper float to sink. It finally closes the second pair of contacts. This trips the circuit breaker or it makes ring an alarm bell for caution. A fault can thus be detected and the transformer is disconnected from the circuit.

e) Transformer Oil

Transformer oil is a mineral oil. It is obtained by refining crude petroleum. It is a good insulator. Its tendency to form, a sludge is very much less. The dielectric strength of oil is affected to a great extent by the presence of moisture. So it should be kept dry. Transformer oil serves two functions,

1. Cooling
2. Insulation

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